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QUANTUM TOPIC MODEL: TOPIC MODELING USING VARIATIONAL QUANTUM CIRCUITS

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ABSTRACT

Quantum machine learning aims to leverage quantum computing to enhance the computing capabilities and storage efficiency of classical machine learning. Among them, quantum generative models, as a type of unsupervised machine learning, have been proven to learn distributions that are outside of classical machine learning reach. However, there is limited work on how to truly utilize and validate a possible quantum advantage in practical applications, especially in the field of natural language processing. This paper proposes the first Quantum Topic Model (QTM) based on variational quantum circuits to validate and showcase the possible advantages of quantum computing for generative models. QTM not only generates random samples like generative models but also exhibits fitting capabilities similar to neural networks. Finally, we conducted experiments on real datasets, and the results showed that QTM achieved a **46.9%** reduction in model parameters while obtaining **10.4%** higher topic coherence and **0.2%** higher topic diversity with fewer iterations than neural topic models.

Index Terms— Quantum machine learning, variational quantum circuit, topic model, natural language processing, and generative model

1. INTRODUCTION

The topic model is a statistical language model capable of modeling and clustering words in massive text data. It is one of the most important techniques in text mining and finds extensive applications in various tasks such as text classification [1], text summarization [2], and community discovery [3]. The essence of topic modeling is to describe the generation process of all words in a document from the perspective of a probabilistic model. The most successful statistical probabilistic model is Latent Dirichlet Allocation (LDA) [4]. Subsequently, a large number of deep generative learning-based topic modeling methods [5, 6, 7] using neural networks (NTM) were proposed to improve the computational speed and modeling capability of LDA. However, due to the deterministic output of neural networks, NTM usually requires complex tricks and pseudo-random number strategies to sample. Additionally, the increasing number of parameters in neural networks also leads to significant resource overhead.

Recently, quantum machine learning has gained attention, particularly the unique advantages of quantum circuits in generative tasks. On one hand, the computational complexity of quantum sampling suggests that quantum circuits have stronger expressiveness than classical neural networks [8]. On the other hand, quantum circuits have been proven to learn distributions beyond the reach of classical models [9]. Additionally, the superposition advantage of qubits [10, 11] suggest that quantum machine learning can express more classical data with fewer parameters, especially in the field of natural language processing. Therefore, we believe that variational quantum circuits may be more suitable for deep generative tasks compared to neural networks. Theoretically, quantum circuits can generate true random samples without any tricks or pseudo-random strategies. Moreover, variational quantum circuits can fit unknown distributions through gradient descent, but with fewer parameters compared to neural networks.

Hence, utilizing quantum generative models for topic modeling is an exciting research problem. It can help us prioritize the verification of the generative role of quantum circuits in the field of natural language processing. To this end, we propose a topic model based on variational quantum circuits (QTM). It employs variational quantum circuits to fit the likelihood function of word distributions and the posterior distribution of topics, respectively. Specifically, we set the prior distribution of topics as a Dirichlet distribution and use Maximum Mean Discrepancy (MMD) [12] as a distribution matching method for the proposal distribution and the prior distribution. Finally, we experimentally verify the effect of QTM. The main contributions of this paper are as follows:

- To the best of our knowledge, QTM is the first work that utilizes quantum circuits for topic modeling.
- QTM is able to outperform the performance of NTM with much fewer parameters in some metrics.

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- QTM may provide a fresh theoretical perspective for the implementation of deep generative learning.

2. RELATED WORK

2.1. Topic Model

LDA [4] is the most classic probabilistic generative model used for topic modeling. It iteratively infers the topics of documents in an interpretable manner through elegant mathematical formulations. However, this model is computationally expensive due to its reliance on variational Bayesian inference or Gibbs sampling. To address this issue, neural topic models (NTM) [13] have been proposed and provide greater flexibility and scalability compared to traditional LDA. One category of such NTMs is based on deep generative networks such as Variational Auto-Encoders (VAE) [5], Generative Adversarial Networks (GAN) [7], or Wasserstein Auto-Encoders (WAE) [6]. These methods employ neural networks to approximate unknown distributions and can explain the generation process of topics similar to LDA. However, due to the deterministic output of neural networks, these models still require some tricks to generate random samples. Furthermore, the fitting ability of neural networks often relies on a large number of parameters, posing a challenge for deep generative models to reduce the model parameters while ensuring the model effectiveness.

2.2. Quantum Natural Language Processing

Quantum natural language processing, as an interdisciplinary field combining quantum machine learning and natural language processing, can be divided into two categories: quantum-inspired (QI) and quantum-generalized (QG) approaches. The QI primarily explores quantum-like phenomena in natural language and has been successfully applied to tasks such as Information Retrieval (IR) [14, 15] and Question Answering (QA) [16, 17]. Unfortunately, they still adhere to classical computation rules in their implementation and have not fully harnessed the practical advantages of quantum machine learning. In contrast, QG primarily explores natural language understanding methods based on quantum computational rules, aiming to find efficient algorithms for parallel text processing and complex semantic computations [10, 11]. The current mainstream trend is to integrate quantum circuits as a model enhancement method into traditional neural networks, such as Long Short Term Memory [18], Convolutional Neural Networks [19], and pre-trained models [20]. However, those quantum models in essence is not the generation model based on unsupervised learning, and they merely demonstrate performance surpassing classical models in specific toys problems rather than practical applications.

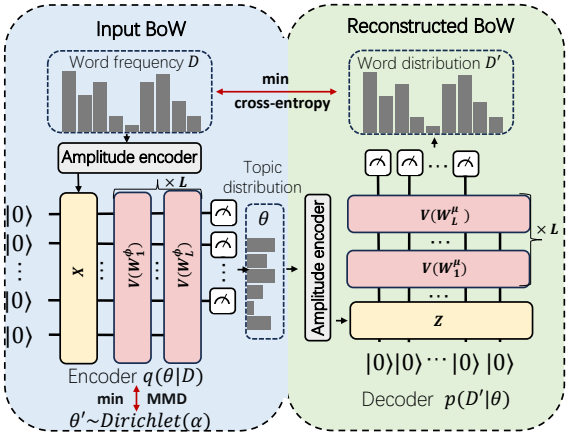


Fig. 1: Architecture of quantum topic model.

3. METHOD

Given a corpus containing N documents $\{D_1, D_2, .., D_N\}$ and a vocabulary with V words $\{w_1, w_2, .., w_V\}$, the task of topic modeling is to extract the topic distributions $\theta \in \mathbb{R}^K$ for every documents, i.e., $\{\theta_1, \theta_2, .., \theta_N\}$ and the word distributions $\beta \in \mathbb{R}^V$ for every topics, i.e., $\{\beta_1, \beta_2, .., \beta_K\}$. Inspired by VAE-based neural topic models [5, 6], in this paper, we propose a quantum version of VAE for topic modeling (QTM), as shown in Fig. 1. Our model can be seen as a quantum generalization of those approach.

3.1. Network Architecture

QTM consists of an encoder $q(\theta|D)$ and a decoder $p(D'|\theta)$, except that the encoder and decoder are replaced with variational quantum circuits. Given a Bag-of-Words (BoW) representation D, the encoder maps it into a K-dimensional probability distribution $\theta \sim p(\theta|D)$:

$$\theta = [|\langle\Theta_1|\phi_W(X)\rangle|^2, |\langle\Theta_2|\phi_W(X)\rangle|^2, ..., |\langle\Theta_K|\phi_W(X)\rangle|^2], \tag{1}$$

where $X = f(D)$ is the amplitude encoding of the BoW D; $p(\theta_{.i}) = |\langle\Theta_i|\phi_W(X)\rangle|^2$ represents the probability of observation to topic i, and $\{\Theta_i\}_{i=1}^K$ is a set of orthonormal bases; $|\phi_W(X)\rangle$ is a wave function with variational parameters W, which reflects the unitary evolution process:

$$|\phi_W(X)\rangle = V(W_L^\phi)...V(W_2^\phi)V(W_1^\phi)|X\rangle. \tag{2}$$

Initially, the quantum state is in the state $|X\rangle$. Through the layers $V(W^\phi)$, the quantum evolution process described in Eq. 2 can be achieved. Then, through probability measurement, as shown in Eq. 1, a normalized topic distribution θ can be obtained. It is noteworthy that on the simulation platforms, the probability distribution of the wave function can be directly obtained, while on the real quantum computers, we should repeatedly sample to estimate the probability distribution.

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Given the topic distribution θ, the decoder uses the variational quantum circuit to output a word distribution $D' \sim p(D'|\theta)$. Similar to the encoder, we also have:

$$D' = [|\langle \mathtt{D}_1|\mu_W(Z)\rangle|^2, |\langle \mathtt{D}_2|\mu_W(Z)\rangle|^2, ..., |\langle \mathtt{D}_V|\mu_W(Z)\rangle|^2], \tag{3}$$

where $Z = f(\theta)$ is the amplitude encoding of the θ; $p(D'_{.i}) = |\langle \mathtt{D}_i|\mu_W(Z)\rangle|^2$ is the probability of observation to word i, and $\{\mathtt{D}_i\}_{i=1}^{V}$ is a set of orthonormal bases; $|\mu_W(Z)\rangle = V(W_L^{\mu})...V(W_2^{\mu})V(W_1^{\mu})|Z\rangle$ is a wave function for decoder.

Then, the reconstruction loss of QTM can be computed using the cross-entropy loss function:

$$\mathcal{C}(D, D') = -\sum_{i=1}^{V} D_{.i} \log(D'_{.i}). \tag{4}$$

Minimizing Eq. 4 allows us to learn the decoder, which effectively maximizes the likelihood probability $p(D'|\theta)$. To further learn the encoder, we need to minimize the distance between the encoder $q(\theta|D)$ and a prior distribution $p(\theta)$. We choose MMD [12] to match distribution:

$$\begin{aligned} \mathrm{MMD}_{\mathbf{k}}\left(q(\theta|D), p(\theta)\right) &= \frac{1}{m(m-1)} \sum_{i \neq j} \mathbf{k}\left(\theta_i, \theta_j\right) \\ &+ \frac{1}{m(m-1)} \sum_{i \neq j} \mathbf{k}\left(\theta'_i, \theta'_j\right) - \frac{2}{m^2} \sum_{i,j} \mathbf{k}\left(\theta_i, \theta'_j\right), \end{aligned} \tag{5}$$

where m is the number of samples. $\theta \sim q(\theta|D)$ and $\theta' \sim p(\theta)$ represent samples from both distributions, respectively. $\mathbf{k}\left(\theta, \theta'\right) = \exp\left(-\arccos^2\left(\sum_{i=1}^{K} \sqrt{\theta_{.i}\theta'_{.i}}\right)\right)$ is a metric kernel. We follow the work [4, 6] and set the prior distribution as the Dirichlet distribution.

Finally, by combining Eq. 4 and Eq. 5, QTM can use gradient descent to learn the optimal parameters. θ and β can be further obtained. We notice that quantum methods exhibit very interesting properties. The output of a quantum circuit is inherently a standard probability distribution without manual normalization and the observation of a quantum circuit is a sampling once. Intuitively, this inherent generated property might be more suitable for modeling probabilistic generative problems than neural networks.

3.2. Circuit Configuration

The variational parameter layer $V(W)$ is the main reason why quantum circuits can fit unknown distributions, and the topology design of the circuit will directly determine the fitting ability of the quantum model. We chose the base entangler shown in Fig. 2. In this circuit, the RY gate is selected as the parametric rotation gate, and the CNOT gate is selected to establish the entanglement correlation between two adjacent qubits. Such variational parameter layers are simple to implement, can be stacked repeatedly, and are easier to simulate on a classical computer.

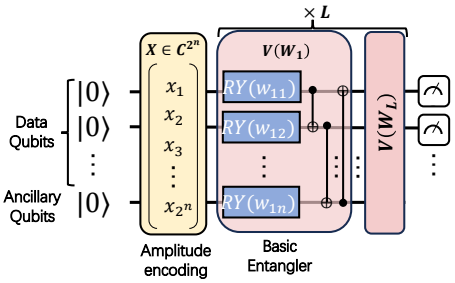


Fig. 2: Variational quantum circuit we designed.

The amplitude coding layer $f(X)$ can map the classical data D to the quantum circuits. Although this method is more efficient than the rotation gate coding, the classical data are usually real numbers and lack imaginary part characteristics, so the amplitude coding cannot fully process the classical data. To this end, we propose a complex-value encoding method: $X = f(D) = X_{real} + \mathtt{i}X_{imag}$, where $X_{real}, X_{imag} \in \mathbb{R}^{2^n}$ are the outputs of two classical linear layers on data D, respectively. $\mathtt{i}$ is the imaginary number with $\mathtt{i}^2 = -1$. Then using normalization $|X\rangle = \frac{X}{||X||}$, $D \in \mathbb{R}^V$ is encoded into $|X\rangle \in \mathbb{C}^{2^n}$ in Hilbert space. n denotes the number of qubits, where $n = \lceil log_2 K \rceil + 1$ for the encoder and $n = \lceil log_2 V \rceil + 1$ for the decoder. It should be noted that this means we can encode a language with a vocabulary of millions of words using only around 20 qubits. In addition, we have also introduced an auxiliary qubit that undergoes evolution without measurement. This is beneficial for avoiding *barren plateaus* [21]. Finally, we design the variational quantum circuit shown in Fig. 2.

4. EXPERIMENTS

4.1. Experimental Setup

Datasets. To verify the performance of QTM, we conduct experiments on two publicly available datasets in different languages as shown in Table 1.

- **20NewsGroup:** 20NewsGroup dataset is one of the international standard datasets for text mining and topic modeling. We use the dataset version from the optimizing and comparing topic models toolkit OCTIS [22].
- **CNews10k:** The CNews10k dataset, provided by the Neural Topic Models toolkit[1], was constructed by selecting Chinese news headlines from THUCNews [23].

Model Setup. We run QTM using the quantum simulation platform provided by PennyLane [24]. It can carry out quantum circuits automatically differential. We set the number of layers $L = 2$ in the quantum circuit, the hyperparameter $\alpha = 1$ of the Dirichlet prior distribution. Finally, the

[1]https://github.com/zll17/Neural_Topic_Models

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Table 1: Details of the datasets. #Docs: Number of documents. #Words: Number of words in vocabulary. Avg. Len: Average document length. Lang.: Document language type.

Datasets	#Docs	#Words	Avg. Len	Lang.
20NewsGroup	16309	1612	48.0	English
CNews10k	9874	3113	18.7	Chinese

model was run for 300 epochs with a learning rate of 1e-2, and we reported the best results obtained during the running process.

Baseline Methods. We selected two representative topic models as baselines:

- **LDA:** LDA [4] is one of the most classical topic models, based on statistical and probabilistic modeling.
- **NTM (W-LDA):** W-LDA [6] based on WAE is a representative work of NTM. QTM can be seen as a quantum generalization version of the method.

In our setting, the encoders of W-LDA consist of multilayer perceptrons, while the decoder is a single-layer classifier. The learning rate is set to 1e-3. The aforementioned models were implemented based on NTMt and iteratively trained for 1,000 epochs. We reported the best results obtained during the running process.

Evaluation Metrics. The performance of topic models can be evaluated by topic diversity (**TD**) [22] and topic coherence (TC), where we choose University of California Irvine method (**UCI**) [25] and Normalised Pointwise Mutual Information (**NPMI**) [26] as the metrics for topic coherence.

4.2. Results and Analysis

We conducted two sets of experiments with the number of topics set to 32 and 64, respectively. The final results are shown in Table 2 and Table 3. For CNews10k, QTM demonstrates overwhelming advantages. Compared to NTM, QTM demonstrates a significant improvement of **9.9%** in UCI and **3.5%** in NPMI when the number of topics is set to 32, while simultaneously achieving a remarkable reduction of the parameter by **72.7%**. Although LDA achieves the highest TD, its topic coherence is inferior to NTM and QTM. When the number of topics is set to 64, QTM delivers the best performance. Compared to NTM, QTM achieves a remarkable improvement of **10.4%** in UCI, **5.3%** in NPMI, and **0.2%** in TP while simultaneously reducing the parameter by an impressive **46.9%**. For dataset 20NewsGroup, QTM is also competitive. When the number of topics is set to 32, compared to LDA and NTM, QTM can improve UCI by 0.6% and TP by 0.4%. When the number of topics is set to 64, QTM achieves the highest UCI and also performs well in NPMI and TD. As 20NewsGroup has a vocabulary closer to the power of 2, the qubit utilization is higher, resulting in a higher percentage of parameter reduction on this dataset.

Table 2: Experimental results on CNews10k.The last line provides the percentage of parameter reduction for QTM compared to NTM. All metrics are better when they are larger.

Model	32 Topics			64 Topics		
	UCI	NPMI	TD	UCI	NPMI	TD
LDA	-12.035	-0.390	**0.979**	-10.923	-0.338	0.745
NTM(W-LDA)	-9.687	-0.268	0.879	-10.776	-0.326	0.748
QTM(Ours)	**-8.726**	**-0.233**	0.833	**-9.649**	**-0.273**	**0.750**
	72.7% Param. ↓			46.9% Param. ↓		

Table 3: Experimental results on 20NewsGroup.

Model	32 Topics			64 Topics		
	UCI	NPMI	TD	UCI	NPMI	TD
LDA	0.468	0.084	0.783	-0.055	0.056	**0.795**
NTM(W-LDA)	0.126	**0.086**	0.743	0.162	**0.056**	0.472
QTM(Ours)	**0.471**	0.081	**0.787**	**0.204**	0.050	0.630
	76.7% Param. ↓			54.7% Param. ↓		

The experimental results are remarkable, as the quantum model achieves comparable or even superior performance to the classical model with the fewer number of iterations and parameters. Firstly, this may be attributed to the inherent non-classical entanglement properties in quantum circuits, which enable the model to uncover hidden patterns that neural networks struggle with. Additionally, quantum circuits possess both distributed learning advantages similar to neural networks and generation characteristics similar to probabilistic generative models, making them feasible for modeling generative models. Finally, the superposition property of quantum circuits enables encoding extensive vocabulary and semantic knowledge with only a few qubits, leading to a significant reduction in parameters in quantum models.

5. CONCLUSION

In this paper, we propose and validate the first quantum topic model QTM. The model follows the architecture of VAE and utilizes variational quantum circuits to fit the likelihood function and posterior distribution. We have demonstrated that QTM can outperform classical methods in certain metrics. This may offer a novel quantum perspective for deep generative learning.

6. ACKNOWLEDGEMENTS

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